4.5.13

EE25BTECH11011 - Navya Banavathu

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Question

Find the equations of the lines parallel to axes and passing through $(2, 3)$

Solution

The given point is

$$A = \begin{pmatrix} 2 \\ 3 \end{pmatrix}. \tag{1}$$

Case 1: Line parallel to the x-axis
A line parallel to the x-axis has direction vector

$$\mathbf{d} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{2}$$

A normal vector to this line is therefore

$$\mathbf{n} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3}$$

Solution

Equation of a line in normal form is

$$\mathbf{n}^T(\mathbf{x} - A) = 0. \tag{4}$$

Substituting,

$$\begin{pmatrix} 0 & 1 \end{pmatrix} \left(\mathbf{x} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) = 0 \tag{5}$$

$$\begin{pmatrix} 0 & 1 \end{pmatrix} \mathbf{x} = 3 \tag{6}$$

Case 2: Line parallel to the y-axis

A line parallel to the y-axis has direction vector

$$\mathbf{d} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{7}$$

A normal vector to this line is therefore

$$\mathbf{n} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{8}$$

Equation of the line:

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \left(\mathbf{x} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) = 0 \tag{9}$$

$$\begin{pmatrix} 1 & 0 \end{pmatrix} \mathbf{x} = 2 \tag{10}$$

Solution

Case 3: General line through $(2,3)$
Let the normal vector be

$$\mathbf{n} = \begin{pmatrix} a \\ b \end{pmatrix}, \quad (a,b) \neq (0,0). \tag{11}$$

Equation:

$$\mathbf{n}^T(\mathbf{x} - A) = 0 \tag{12}$$

$$\begin{pmatrix} a & b \end{pmatrix} \left(\mathbf{x} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right) = 0 \tag{13}$$

$$\begin{pmatrix} a & b \end{pmatrix} \mathbf{x} = 2a + 3b \tag{14}$$

Code (C)

Part 1

```c
#include <stdio.h>

int main(void) {
    double Ax = 2.0, Ay = 3.0;
    printf("A = (%.1f, %.1f)\n\n", Ax, Ay);

    double a1 = 0.0, b1 = 1.0, rhs1 = 3.0;
    printf("Case 1 (parallel to x-axis):\n");
    printf("[%.0f %.0f]  x = %.0f\n\n", a1, b1, rhs1);
```

Code (C)

Part 2

```c
double a2 = 1.0, b2 = 0.0, rhs2 = 2.0;
   printf("Case 2 (parallel to y-axis):\n");
   printf("[%.0f %.0f]  x = %.0f\n\n", a2, b2, rhs2);

   double a = 4.0, b = -1.0;
   double rhs = Ax * a + Ay * b;
   printf("General case (a=%.0f, b=%.0f):\n", a, b);
   printf("[%.0f %.0f]  x = %.0f\n", a, b, rhs);

   return 0;
}
```

Python Code:Import and Define points

Part 1

```
import ctypes
import matplotlib.pyplot as plt
import numpy as np
import platform

# Load compiled C library
if platform.system() == "Windows":
    lib = ctypes.CDLL("./code6.dll")
else:
    lib = ctypes.CDLL("./libcode6.so")

# Tell ctypes about function signatures
lib.get_x_line.argtypes = [ctypes.c_double, ctypes.c_double]
lib.get_x_line.restype = ctypes.c_double

lib.get_y_line.argtypes = [ctypes.c_double, ctypes.c_double]
lib.get_y_line.restype = ctypes.c_double
```

Python Code

Part 2

```
lib.get_general_line.argtypes=[ctypes.c_double, ctypes.c_double,
                               ctypes.c_double, ctypes.c_double]
lib.get_general_line.restype = ctypes.c_double

# Point A
Ax, Ay = 2.0, 3.0

# Case 1: Line parallel to x-axis (n = (0,1))
y_const = lib.get_y_line(Ax, Ay)

# Case 2: Line parallel to y-axis (n = (1,0))
x_const = lib.get_x_line(Ax, Ay)

# Case 3: General line (example n = (2,-1))
n1, n2 = 2.0, -1.0
rhs = lib.get_general_line(Ax, Ay, n1, n2)
```

Python Code: Plotting

Part 3

```
fig, ax = plt.subplots(figsize=(7,7))

x_vals = np.linspace(-5, 10, 400)
y_vals_range = np.linspace(-5, 10, 400)

# Case 1: horizontal line
ax.plot(x_vals, y_const * np.ones_like(x_vals), 'r--',
        label=rf"$(0,1)\cdot x = {y_const:.1f}$ (Line parallel to
            x-axis)")

# Case 2: vertical line
ax.plot(x_const * np.ones_like(y_vals_range), y_vals_range,'b--',
        label=rf"$(1,0)\cdot x = {x_const:.1f}$ (Line parallel to
            y-axis)")

# Case 3: general line (nx = rhs)
y_vals = (rhs - n1*x_vals) / n2
```

Python Code: Plotting

Part 4

```
ax.plot(x_vals, y_vals, 'g-',
        label=rf"$({n1:.0f},{n2:.0f})\cdot x = {rhs:.1f}$ (
            Equation of line passing through (2,3))")

# Mark point A
ax.scatter(Ax, Ay, color='black', zorder=5)
ax.text(Ax+0.1, Ay+0.1, r"$A(2,3)$")

ax.set_xlim(-5, 10)
ax.set_ylim(-5, 10)
ax.set_xlabel("x")
ax.set_ylabel("y")
ax.legend()
ax.grid(True)
ax.set_aspect('equal', adjustable='box')
plt.savefig("fig6.png")
plt.show()
```

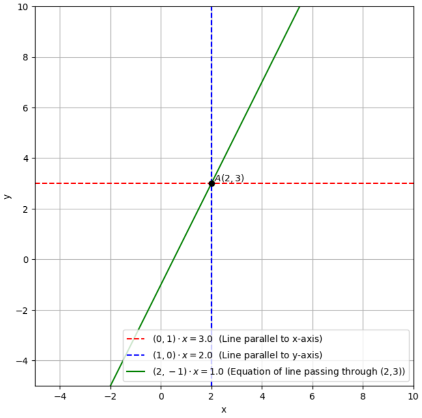

Figure: Equations of line parallel to x,y axes and passing through(2,3)